macOS Automated Device Enrollment with SSO

Implementation Solution Guide

Version 1.0.0 · 2026-06-14

Contents

Executive Summary — macOS Automated Device Enrollment with SSO 4

1. Solution Overview — macOS Automated Device Enrollment with SSO 4

What this solution is . . . 4

How it works . . . 5

Platform SSO authentication methods (choose one) . . . 5

Benefits . . . 5

Limitations (be honest with buyers) . . . 5

The standard deployment model . . . 6

2. Requirements — macOS Automated Device Enrollment with SSO 6

R1. Apple Business Manager (or Apple School Manager) account . . . 6

R2. Apple Push Notification service (APNs) certificate . . . 6

R3. Automated Device Enrollment token (ABM ↔ Intune connection) . . . 7

R4. Devices present in Apple Business Manager . . . 7

R5. Devices assigned to the Intune MDM server . . . 7

R6. Supported macOS version . . . 7

R7. Company Portal app for macOS . . . 7

R8. Microsoft Enterprise SSO extension capability . . . 7

R9. Network access to Apple management endpoints . . . 8

R10. macOS enrollment enabled in Intune . . . 8

Global Requirements — Intune Solutions (category-level) 8

G1. Microsoft Entra ID tenant . . . 8

G2. Microsoft Intune subscription . . . 8

G3. Administrative roles . . . 8

G4. Device enrollment in Intune . . . 9

G5. Pilot security group . . . 9

G6. Supported, current operating system . . . 9

G7. Network access to required cloud endpoints . . . 9

3. Implementation — macOS Automated Device Enrollment with SSO 9

Phase 1 — Connect Apple to Intune . . . 10

Phase 2 — Assign devices and define the enrollment experience . . . 10

Phase 3 — Deploy Company Portal and Platform SSO . . . 11

Phase 4 — Pilot, validate, expand . . . 12

5. Testing & Validation — macOS Automated Device Enrollment with SSO 13

Test 1 — Tenant readiness . . . 13

Test 2 — Device appears as ADE . . . 13

Test 3 — Zero-touch enrollment . . . 13
Test 4 — Supervision and lock . . . 13
Test 5 — Company Portal + Platform SSO registration . . . 13
Test 6 — Single sign-on works . . . 14
Test 7 — Secure Enclave authentication method . . . 14
Test 8 — Compliance & Conditional Access . . . 14
Acceptance gate . . . 14
6. Troubleshooting — macOS Automated Device Enrollment with SSO 14
The Mac never shows a "Remote Management" screen . . . 14
Enrollment fails partway / device not in Intune . . . 15
Device enrolled but **not Supervised** . . . 15
Company Portal doesn't install . . . 15
Platform SSO never prompts to register . . . 15
Registered, but apps still prompt for a password . . . 15
Secure Enclave method not taking effect . . . 15
"Awaiting configuration" / device stuck at Setup Assistant . . . 16
Token / certificate expiry (preventive) . . . 16
Useful checkpoints . . . 16
7. Best Practices — macOS Automated Device Enrollment with SSO 16
Identity & authentication . . . 16
Enrollment hygiene . . . 16
Token & certificate lifecycle (the #1 cause of outages) . . . 17
Operating system & updates . . . 17
Rollout discipline . . . 17
Operations & documentation . . . 17
Security posture . . . 17
8. Deliverables Summary — macOS Automated Device Enrollment with SSO 18
Documentation (this guide) . . . 18
PowerShell scripts (Microsoft Graph) — `ADEMAC-Scripts.zip` . . . 18
Hero image . . . 19
How it's delivered . . . 19
Appendix A — PowerShell Scripts 20

Executive Summary — macOS Automated Device Enrollment with SSO

Apple **Automated Device Enrollment (ADE)** turns every company-owned Mac into a zero-touch, supervised, fully managed device the moment it is powered on — no manual setup, no local admin hand-holding, no imaging. Paired with **Platform Single Sign-On (Platform SSO)** and the **Microsoft Enterprise SSO** plug-in, your users sign in to their Mac with their **Microsoft Entra ID** credentials and carry that identity, seamlessly and securely, into every app and browser.

This solution delivers the complete, field-tested method to stand this up in **Microsoft Intune** — written so a single administrator can follow it end to end:

• **Zero-touch provisioning.** Macs bought through Apple Business Manager (or added with Apple

Configurator) enroll automatically, supervised, with your branding and your apps, straight out of the box.

• **Entra-backed sign-in.** Platform SSO registers the Mac with Microsoft Entra ID and lets users

unlock the device and authenticate to Microsoft 365 and SSO-enabled apps using their corporate identity — with a hardware-bound **Secure Enclave** key, not a synced password.

• **Conditional Access ready.** Enrolled, compliant, Entra-registered Macs become first-class

citizens in your Zero Trust model.

The package follows the standard repeatable shape — **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** — with exact, click-by-click Intune steps, a requirements guide that tells you how to verify *and* fix each prerequisite, five ready-to-run PowerShell scripts (Microsoft Graph), and testing/troubleshooting guides with clear pass/fail gates.

Outcome: company Macs that deploy themselves, authenticate with corporate identity, and meet compliance — delivered by your own team, in a controlled pilot, without a consulting engagement.

1. Solution Overview — macOS Automated Device Enrollment with SSO

What this solution is

This solution combines two Apple + Microsoft capabilities into one repeatable deployment:

1. **Automated Device Enrollment (ADE)** — Apple's zero-touch enrollment program (formerly DEP).

Company-owned Macs that belong to your **Apple Business Manager (ABM)** organization are automatically enrolled into **Microsoft Intune**, **supervised**, the first time they run Setup Assistant. Management is mandatory and locked to the device.

1. **Platform Single Sign-On (Platform SSO)** — Apple's macOS framework that lets a third-party

identity provider extend sign-in to the Mac itself. With the **Microsoft Enterprise SSO** extension, the Mac registers with **Microsoft Entra ID** and users authenticate with their corporate identity at the login window and across apps and Safari/Edge/Chrome.

Together they give you a Mac that provisions itself and signs in with corporate identity — managed, supervised, and Conditional-Access ready.

How it works

1. **Apple Business Manager** → **Intune.** You connect ABM to Intune with an **Automated Device Enrollment token** (a server token). Macs assigned to the Intune MDM server in ABM appear in Intune as ADE devices.

1. **Enrollment profile.** An **enrollment program profile** in Intune defines the out-of-box experience: supervised mode, user affinity, which Setup Assistant screens to skip, and whether management is locked.

1. **Power on** → **enroll.** When a user (or technician) powers on the Mac and connects to a network, Setup Assistant contacts Apple, discovers the Intune MDM server, and enrolls automatically. The device becomes supervised and managed; your apps and policies flow down.

1. **Platform SSO registration.** A configuration profile deploys the **Microsoft Enterprise SSO** extension. On first sign-in, the **Company Portal** app completes Platform SSO **registration**: the Mac generates a hardware-bound key in the **Secure Enclave** and registers it with Entra ID.

1. **Single sign-on.** From then on, the user's macOS sign-in is tied to their Entra identity, and apps/browsers that support the Microsoft identity broker get silent SSO — no repeated passwords.

Platform SSO authentication methods (choose one)

• **Secure Enclave key (recommended)** — a hardware-bound key in the Mac's Secure Enclave signs authentication. Phishing-resistant; the local password is decoupled from Entra. Best security.

• **Password sync** — the user's local macOS password is kept in sync with their Entra password. Simplest for users; least hardened.

• **Smart card** — authentication via a smart card/certificate. For high-assurance environments.

This solution standardizes on the **Secure Enclave key** method, with notes for the alternatives.

Benefits

• **Zero-touch deployment** — Macs enroll supervised on first boot; no imaging, no manual MDM steps.

• **Corporate identity on the Mac** — Entra sign-in at the login window and across apps/browsers.

• **Phishing-resistant** — Secure Enclave key, hardware-bound, never leaves the device.

• **Supervised management** — stronger controls, app management, and wipe protection.

• **Conditional Access** — enrolled, compliant, Entra-registered Macs fit your Zero Trust model.

• **Better experience** — users sign in once with the identity they already know.

Limitations (be honest with buyers)

• ADE requires devices to be in **Apple Business Manager** (purchased via Apple/authorized reseller, or added manually with **Apple Configurator** — the latter adds a provisional 30-day release period).

• Platform SSO requires **macOS 13 Ventura or later** (macOS 14 Sonoma+ strongly recommended for the best registration and Secure Enclave experience).

• Requires an **Apple Push Notification service (APNs) certificate**, renewed annually, and an **ADE/ABM token**, renewed annually — both are hard expiry dates that block enrollment if missed.

• Platform SSO registration depends on the **Company Portal** app and the **Microsoft Enterprise SSO** extension being present and current.

• Personally-owned (BYO) Macs use a different (user-enrollment) path and are out of scope here.

The standard deployment model

This solution uses **one** model so the process is repeatable across our catalog:

> ***Company-owned, Apple Business Manager–sourced Macs, automatically enrolled into Intune (supervised), with Platform SSO via the Microsoft Enterprise SSO extension and a Secure Enclave key.***

Every step follows the repeatable pattern: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

2. Requirements — macOS Automated Device Enrollment with SSO

Confirm **every** requirement below before you begin. Each entry follows the same pattern: **Description · Why required · How to verify · How to implement if missing.** Category-wide prerequisites (tenant, Intune licensing, admin roles, pilot group, network) are covered once in the **Global Requirements — Intune Solutions** document; the items here are specific to Apple ADE + Platform SSO.

R1. Apple Business Manager (or Apple School Manager) account

Description: A verified Apple Business Manager (ABM) organization. **Why:** ADE only works for devices that belong to your ABM organization; ABM is also where you assign devices to the Intune MDM server and accept device-management terms. **Verify:** Sign in to `https://business.apple.com` with an ABM Administrator account; the organization status is **active**. **Implement:** Enroll at `https://business.apple.com/enroll` using your **D-U-N-S** number; complete Apple's verification (a phone callback to a verifiable company contact). Allow 1–5 business days.

R2. Apple Push Notification service (APNs) certificate

Description: A valid MDM Push certificate connecting Intune to Apple's push service. **Why:** Every Apple device managed by Intune (iOS/iPadOS/macOS) requires APNs; without it, no Mac can enroll or receive commands. **Verify:** Intune admin center → **Devices** → **Enrollment** → **Apple** → **Apple MDM Push certificate** shows **Active** with a future expiration date. **Implement:** In Intune, download the CSR, sign in to

`https://identity.apple.com` with a **company Apple ID** (not a personal one), create/renew the certificate, and upload the resulting `.pem`. **Renew annually with the same Apple ID** — a new Apple ID re-issues the cert and unenrolls all devices.

R3. Automated Device Enrollment token (ABM ↔ Intune connection)

Description: A downloaded ABM server token (`.p7m`) uploaded into Intune as an enrollment program token. **Why:** This is the trust link that lets ABM-assigned devices enroll into your Intune tenant. **Verify:** Intune → **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens**; a token is listed with a future expiration. **Implement:** In Intune, download the public key; in ABM **Preferences** → **MDM Servers**, create an MDM server, upload the public key, download the server token; upload that token back into Intune. **Renew annually.**

R4. Devices present in Apple Business Manager

Description: The target Macs appear in ABM and are assignable to the Intune MDM server. **Why:** Only ABM devices can use ADE; a Mac not in ABM cannot be auto-enrolled. **Verify:** ABM → **Devices**; the Mac's serial number is listed. **Implement:** Purchase through Apple or an **Apple-authorized reseller** linked to your ABM (devices appear automatically), or add existing Macs with **Apple Configurator** (note: Configurator-added devices get a **30-day provisional period** during which the user can release them from management).

R5. Devices assigned to the Intune MDM server

Description: Each target Mac is assigned to the Intune MDM server in ABM. **Why:** Assignment tells Apple which MDM a device should enroll into during Setup Assistant. **Verify:** ABM → **Devices** → select the device → **MDM Server** shows your Intune server. **Implement:** ABM → **Devices** → select devices → **Edit MDM Server** → choose the Intune MDM server. Then in Intune, **Sync** the enrollment token.

R6. Supported macOS version

Description: macOS **13 Ventura or later** (macOS **14 Sonoma+ recommended**). **Why:** Platform SSO and the Secure Enclave authentication method require macOS 13+; later releases materially improve registration reliability. **Verify:** Apple menu → **About This Mac**, or Intune → device → **Hardware**. **Implement:** Upgrade via **System Settings** → **General** → **Software Update**, or deploy a minimum-OS **compliance policy** so non-compliant Macs are flagged and updated.

R7. Company Portal app for macOS

Description: The Intune **Company Portal** application installed on the Mac. **Why:** Company Portal performs the Platform SSO **registration** with Microsoft Entra ID and reports device state; Platform SSO does not complete without it. **Verify:** Intune → **Apps** → **macOS**; "Company Portal" is added and assigned to your pilot group. **Implement:** Add Company Portal (Intune → Apps → macOS → Add → choose the macOS Company Portal) and assign it **Required** to the pilot group so it installs automatically post-enrollment.

R8. Microsoft Enterprise SSO extension capability

Description: The **Microsoft Enterprise SSO** plug-in delivered via a macOS **SSO app extension / Platform SSO** configuration profile. **Why:** This Apple-approved redirect extension is what brokers Entra authentication for Platform SSO and app/browser single sign-on. **Verify:** Intune → **Devices** → **Configuration**; a profile of type **SSO app extension** (Platform SSO) exists targeting your pilot group. **Implement:** Create it during deployment (Implementation Guide §6), or run the included `Set-PlatformSSOProfile.ps1` to provision it via Microsoft Graph.

R9. Network access to Apple management endpoints

Description: Outbound HTTPS to Apple's enrollment, push, and software endpoints. **Why:** Setup Assistant, ADE discovery, APNs, and app delivery all depend on Apple endpoints. **Verify:** From the device network, reach `https://mdmenrollment.apple.com`, `https://gdmf.apple.com`, and the Apple push range `*.push.apple.com` (TCP 443 and 5223). **Implement:** Allow Apple's published network ranges (Apple support article HT210060) plus the Microsoft Intune/Entra endpoints through proxy/firewall. Do **not** TLS-inspect Apple push/enrollment.

R10. macOS enrollment enabled in Intune

Description: macOS platform enrollment is permitted by enrollment restrictions. **Why:** A restriction that blocks macOS will silently stop ADE devices from enrolling. **Verify:** Intune → **Devices** → **Enrollment** → **Enrollment device platform restrictions** → **macOS**; platform is **Allowed**. **Implement:** Edit the default (or a targeted) restriction to **Allow** macOS for the pilot group.

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level on the marketplace so individual solution listings stay small. Confirm these are in place before deploying *any* Intune solution.

Each entry: **Description · Why required · How to verify · How to implement if missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, enrollment, and security policies. **Verify:** `https://intune.microsoft.com` loads and **Tenant administration** → **Tenant status** shows an active subscription. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies, enrollment connectors, and configuration profiles requires elevated rights. **Verify:** Entra → **Roles and**

administrators; confirm **Intune Administrator** (or Global Administrator for connector setup). **Implement:** Assign **Intune Administrator** to the operator; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted devices are enrolled and manageable in Intune. **Why:** Policies only reach enrolled devices, regardless of platform (Windows, macOS, iOS/iPadOS, Android). **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Configure the appropriate enrollment method for the platform (e.g. automatic enrollment for Windows, Automated Device Enrollment for Apple) and enroll the device.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted to a small group before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists with intended members. **Implement:** Entra → **Groups** → **New group** → type **Security** → add pilot members.

G6. Supported, current operating system

Description: A supported OS baseline with current updates for the targeted platform. **Why:** Modern Intune features and security settings assume a patched, vendor-supported OS. **Verify:** Check the device's OS version (e.g. `winver` on Windows, **About This Mac** on macOS) or Intune → device → **Hardware**. **Implement:** Deploy update policies (Windows Update rings, macOS update/compliance policies) or upgrade to a supported release.

G7. Network access to required cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs (plus Apple endpoints for Apple-platform solutions). **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges (and Apple's published ranges where applicable) through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation — macOS Automated Device Enrollment with SSO

Follow these steps **in order**. The *why* for each capability lives in the Solution Overview; the steps below are deliberately short and exact so you can **click to implement**. Throughout, the repeatable pattern applies: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.** Do everything against a **pilot group** first.

All admin-center steps use the Microsoft Intune admin center at `https://intune.microsoft.com` and Apple Business Manager at `https://business.apple.com` unless noted.

Phase 1 — Connect Apple to Intune

Step 1. Create the Apple MDM Push (APNs) certificate

1. In Intune, go to **Devices** → **Enrollment** → **Apple**.
2. Click **Apple MDM Push certificate**.
3. Check **I agree**, then **Download** the CSR file to your computer.
4. Click **Create your MDM push certificate** to open `https://identity.apple.com`.
5. Sign in with a **company Apple ID** (record which Apple ID you use — you must reuse it every year).
6. Click **Create a Certificate**, upload the CSR, then **Download** the `.pem` certificate.
7. Back in Intune, enter that **Apple ID** in the box, **Browse** to the `.pem`, and click **Upload**.
8. Confirm the status shows **Active** with an expiration ~1 year out.

Step 2. Connect Apple Business Manager (download the Intune public key)

1. In Intune, go to **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens**.
2. Click **Add**.
3. Check **I agree**, then **Download** the Intune **public key** (`.pem`).

Step 3. Create the MDM server in Apple Business Manager

1. In ABM, go to **Preferences (your name, bottom-left)** → **MDM Servers**.
2. Click **Add** (or the **+**). Name it clearly, e.g. **Intune – Production**.
3. Under **MDM Server Settings**, **Upload** the Intune public key from Step 2.
4. Click **Save**, then **Download Token** — this downloads the server token (`.p7m`).

Step 4. Upload the ABM token into Intune

1. Back on the Intune **Add enrollment program token** page, enter the **Apple ID** used for ABM.
2. **Browse** to the `.p7m` token file and **Upload**.
3. The token appears under **Enrollment program tokens** with a future expiration.

Phase 2 — Assign devices and define the enrollment experience

Step 5. Assign Macs to the Intune MDM server (in ABM)

1. In ABM, go to **Devices**.
2. Select the target Mac(s) by serial number (or filter by purchase/order).
3. Click **Edit MDM Server** → choose your **Intune – Production** server → **Continue** → **Confirm**.

Step 6. Sync and confirm devices in Intune

1. In Intune, open **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens** → **(your token)** → **Devices**.
2. Click **Sync** (Apple rate-limits syncs; allow a few minutes).
3. Confirm the assigned serial numbers appear.

Step 7. Create the macOS enrollment program profile

1. On the token, open **Profiles** → **Create profile** → **macOS**.

2. **Basics:** name it (e.g. **macOS ADE – Pilot**); set **User affinity** = **Enroll with user affinity** (required for Platform SSO so the device ties to a user).

1. **Management Settings:**

• **Device management options** → enable **Supervised**.

• **Locked enrollment** → **Yes** (prevents users from removing management).

• **Sync with Company Portal** / **Install Company Portal** as prompted (so SSO registration can run).

1. **Setup Assistant:** hide unnecessary screens (Restore, Apple ID, Siri, Analytics, Touch ID setup as your policy dictates) for a clean first-run; keep **Location** and **Privacy** as required.

1. **Review + Create.**

Step 8. Assign the enrollment profile to the token's devices

1. On the token → **Devices**, select the pilot serial numbers.

2. Click **Assign profile** → choose **macOS ADE – Pilot** → **Assign**.

Step 9. Confirm macOS enrollment is allowed

1. Go to **Devices** → **Enrollment** → **Enrollment device platform restrictions**.

2. Open the default (or pilot-targeted) restriction → **macOS** → ensure **Allow**.

Phase 3 — Deploy Company Portal and Platform SSO

Step 10. Add and assign the Company Portal app

1. Go to **Apps** → **macOS** → **Add**.

2. Choose **macOS** app type for **Company Portal** (or add it as a managed app), **Select**.

3. Assign it **Required** to your **pilot group** so it installs automatically after enrollment.

Step 11. Create the Platform SSO configuration profile

1. Go to **Devices** → **Configuration** → **Create** → **New policy**.

2. **Platform** = **macOS**; **Profile type** = **Settings catalog** → **Create**.

3. Name it **macOS Platform SSO – Pilot**.

4. Click **Add settings**; search **Single Sign On Extension** and add the **Extensible Single Sign-On (SSO)** settings.

1. Configure:

• **Extension Identifier:** `com.microsoft.CompanyPortalMac.ssoextension`

• **Team Identifier:** `UBF8T346G9`

- **Type: Redirect**
- **URLs:** `https://login.microsoftonline.com`, `https://login.microsoft.com`, `https://sts.windows.net`, `https://login.partner.microsoftonline.cn`, `https://login.chinacloudapi.cn`, `https://login.microsoftonline.us`, `https://login-us.microsoftonline.com`
- **Platform SSO** → **Authentication Method: Secure Enclave Key** (recommended).
- **Registration token:** `{{DEVICEREGISTRATION}}` (enables silent device registration).

1. **Review + Create**, then **Assign** to the pilot group.

Prefer automation? Run the included `Set-PlatformSSOProfile.ps1` (Microsoft Graph) to create and assign this profile with the values above — then verify it in the console.

Step 12. (Hybrid/optional) Conditional Access for managed Macs

1. Go to **Entra** → **Protection** → **Conditional Access**.
2. Create a policy (**name it** `CA – macOS Compliant – Pilot`) to require **device compliance** (and/or Entra registration) for macOS accessing Microsoft 365 — **start in report-only mode** for the pilot.

Phase 4 — Pilot, validate, expand

Step 13. Pilot-enroll a test Mac

1. **Erase** the pilot Mac (**System Settings → General → Transfer or Reset → Erase All Content and Settings**) so it returns to Setup Assistant. *(For a Configurator-added device, complete this before the 30-day provisional period ends.)*
1. Power on, choose language/region, connect to network.
2. Setup Assistant shows a **Remote Management** screen confirming your organization manages the Mac.
3. Sign in with the pilot user's **Entra ID** credentials when prompted; complete enrollment.
4. After desktop loads, **Company Portal** installs and prompts to **Register** — complete Platform SSO registration (a small key icon / login-window prompt confirms it).

Step 14. Validate

1. Run `ADEMAC-Discovery.ps1` (tenant readiness) and `ADEMAC-Validation.ps1` (this device): confirm the Mac is **enrolled, supervised, ADE**, has the **SSO profile applied**, and is **registered** with Entra.
1. On the Mac, open a Microsoft 365 web app and confirm **silent SSO** (no password prompt).
2. Confirm compliance in **Intune** → **Devices** → **macOS** → **(device)** → **Compliance**.

Step 15. Expand

1. Once the pilot passes, assign the enrollment profile and the Company Portal/SSO/compliance

policies to broader groups (or make the enrollment profile the **default** for the token).

1. Move any Conditional Access policy from **report-only** to **on** for macOS.
2. Roll out in waves, validating each wave with the scripts before widening.

5. Testing & Validation — macOS Automated Device Enrollment with SSO

Validate against a **pilot device** before expanding. Each test has an explicit **pass/fail gate**. Use the included PowerShell scripts (Microsoft Graph) for tenant-side checks and the on-device steps for the user experience.

Test 1 — Tenant readiness

Procedure: Run `ADEMAC-Discovery.ps1`. **Pass:** APNs certificate **Active** (expiry > 30 days), an **ADE/enrollment token** present (expiry > 30 days), at least one **macOS enrollment profile**, Company Portal app assigned, and a **Platform SSO** configuration profile present. **Fail:** Any of the above missing or expiring within 30 days → resolve before enrolling devices.

Test 2 — Device appears as ADE

Procedure: In Intune → Enrollment program tokens → (token) → **Devices**, locate the pilot serial; or run `ADEMAC-DeviceReport.ps1`. **Pass:** The serial is listed, **profile assigned**, status ready for enrollment. **Fail:** Serial missing → re-check ABM MDM-server assignment (R5) and **Sync**.

Test 3 — Zero-touch enrollment

Procedure: Erase the pilot Mac and run Setup Assistant on a managed network. **Pass:** A **Remote Management** screen names your organization; enrollment completes without manual MDM profile installation; the device shows in **Intune** → **Devices** → **macOS** within minutes. **Fail:** No Remote Management screen → device not assigned/synced, or network blocks `mdmenrollment.apple.com` (R9).

Test 4 — Supervision and lock

Procedure: Inspect the enrolled device in Intune (or run `ADEMAC-Validation.ps1`). **Pass:** Device is **Supervised = Yes** and **Corporate-owned**; management is **locked** (user cannot remove the MDM profile). **Fail:** Supervised = No → the enrollment profile did not enable Supervised (Step 7); re-erase and re-enroll after correcting.

Test 5 — Company Portal + Platform SSO registration

Procedure: After desktop loads, confirm Company Portal installed and complete the **Register** prompt; then check the SSO state. **Pass:** Company Portal reports the device **registered**; macOS shows the Platform SSO login-window prompt / key indicator; `ADEMAC-Validation.ps1` reports the SSO profile **applied**. **Fail:** No registration prompt → confirm Company Portal is **Required** (R7) and the SSO profile is assigned to the user/device (Step 11), then retry.

Test 6 — Single sign-on works

Procedure: On the Mac, open a Microsoft 365 web app (e.g. `https://outlook.office.com`) in Safari and Edge/Chrome; open a native app that supports the broker. **Pass:** Sign-in is **silent** (no password prompt) using the registered Entra identity. **Fail:** Repeated password prompts → verify the SSO extension URLs and identifiers (Step 11), and that the Microsoft Enterprise SSO extension is present.

Test 7 — Secure Enclave authentication method

Procedure: Confirm the Platform SSO method in effect. **Pass:** Authentication uses the **Secure Enclave key** (hardware-bound); the user's local password is decoupled from Entra. **Fail:** Method shows password sync when Secure Enclave was intended → correct the SSO profile setting (Step 11) and re-register.

Test 8 — Compliance & Conditional Access

Procedure: Review the device's **Compliance** in Intune; if Conditional Access is configured, test access from the Mac. **Pass:** Device is **Compliant**; Conditional Access (report-only or on) evaluates the Mac as a managed, registered device. **Fail:** Non-compliant → review the assigned compliance policy and OS minimum (R6).

Acceptance gate

Proceed to **Expand** only when Tests 1–8 pass on at least one pilot Mac and the user-experience checks (silent SSO, clean first-run) are confirmed by a real pilot user.

6. Troubleshooting — macOS Automated Device Enrollment with SSO

Work top-down: confirm the **connection** (APNs/token), then **assignment** (ABM → Intune), then the **device** (enrollment), then **Platform SSO**. Most failures are an expired token, an unassigned device, or a blocked network endpoint.

The Mac never shows a "Remote Management" screen

- **Device not assigned to the Intune MDM server.** ABM → **Devices** → confirm **MDM Server** = your Intune server (R5). After fixing, Intune → token → **Sync**.
- **Profile not assigned.** Intune → token → **Devices** → select serial → **Assign profile**.
- **Already past Setup Assistant.** ADE only triggers during Setup Assistant — **Erase All Content and Settings** and try again.
- **Network blocks Apple.** Ensure `mdmenrollment.apple.com` and `gdmf.apple.com` are reachable and **not** TLS-inspected (R9).
- **Configurator device past 30-day window** or released by the user — re-add and re-assign.

Enrollment fails partway / device not in Intune

- **APNs certificate invalid/expired.** Intune → Apple MDM Push certificate must be **Active** (R2).

If it was renewed with a *different* Apple ID, devices unenroll — restore the original Apple ID.

- **Enrollment token expired.** Renew the ABM server token (R3); re-upload to Intune.
- **macOS blocked by enrollment restriction.** Allow macOS for the target group (R10).

Device enrolled but not Supervised

- The enrollment profile didn't enable **Supervised** (Step 7). Correct the profile, then **erase and re-enroll** — supervision is set at enrollment and cannot be toggled on an enrolled device.

Company Portal doesn't install

- Confirm the app is **Required** and assigned to the **user/group** that owns the device (R7).
- Force a check-in: on the Mac, open **Company Portal** (if present) or **System Settings → Privacy & Security → Profiles, and confirm management; in Intune, use** Sync** on the device.

Platform SSO never prompts to register

- **Company Portal missing or outdated** — it performs registration (R7); update it.
- **SSO profile not assigned** to the signed-in user (Step 11). Confirm assignment and that the enrollment used **user affinity** (Step 7).
- **Extension identifiers wrong** — must be `com.microsoft.CompanyPortalMac.ssoextension` / team `UBF8T346G9` (Step 11).
- **macOS too old** — Platform SSO requires macOS 13+ (14+ recommended) (R6).
- After fixing, sign out/in or reboot to re-trigger the registration prompt.

Registered, but apps still prompt for a password

- **Missing SSO URLs** — ensure all Microsoft login URLs are listed in the SSO extension (Step 11).
- **Browser without broker support** — Safari/Edge/Chrome with the Microsoft SSO extension get silent SSO; some third-party browsers won't.
- **Conditional Access interruption** — a policy may require interaction; review CA sign-in logs in Entra.

Secure Enclave method not taking effect

- The profile may be set to **password sync**. Set **Authentication Method = Secure Enclave Key**

(Step 11), re-assign, and **re-register** (sign out/in). Existing registrations keep their original method until re-registered.

"Awaiting configuration" / device stuck at Setup Assistant

- Usually a transient Apple/Intune sync delay or a network hiccup. Confirm endpoints (R9), **Sync** the token, and allow several minutes; if persistent, erase and retry.

Token / certificate expiry (preventive)

- Both the **APNs certificate** and the **ADE token** expire **annually** and **block enrollment** if lapsed. Run `ADEMAC-Discovery.ps1` (or `ADEMAC-Remediation.ps1`) regularly; it flags anything expiring within 30 days. Renew with the **same Apple ID** each time.

Useful checkpoints

- **Intune:** Devices → macOS → (device) → **Hardware / Compliance / Configuration**; Enrollment → Apple → token → **Devices**.
- **On the Mac: System Settings** → **Privacy & Security** → **Profiles** (MDM profile present); **Console.app** filtered for `AppSSO` / `DeviceManagement` for SSO and enrollment logs.
- **Scripts:** re-run `ADEMAC-Validation.ps1` after each change to confirm the gate.

7. Best Practices — macOS Automated Device Enrollment with SSO

Field-proven guidance to keep the deployment reliable, secure, and low-maintenance.

Identity & authentication

- **Standardize on the Secure Enclave key method.** It is phishing-resistant and decouples the local password from Entra. Reserve password sync for edge cases that genuinely need it.
- **Require user affinity** on the enrollment profile so each Mac is tied to a user and Platform SSO can register. Device-affinity (shared) Macs need a different design.
- **Layer Conditional Access** to require compliant, registered Macs for Microsoft 365 — start in **report-only**, then enforce once the pilot is clean.

Enrollment hygiene

- **Always supervise** company Macs (Step 7) and **lock enrollment** so management can't be removed.
- **Buy through channels linked to ABM** so devices appear automatically; use Apple Configurator only for legacy stock, and complete enrollment before the **30-day provisional period** lapses.

• **Keep one well-named MDM server** per environment in ABM (e.g. *Intune – Production*) to avoid assigning devices to the wrong tenant.

Token & certificate lifecycle (the #1 cause of outages)

• **Track both annual expiries** — the **APNs certificate** and the **ADE token**. A lapse blocks all new enrollments and can unenroll devices.

• **Always renew APNs with the same company Apple ID.** Document which Apple ID owns it; never use a personal Apple ID or one tied to an individual who may leave.

• **Schedule** `ADEMAC-Discovery.ps1` to run regularly and alert on anything within 30 days of expiry.

Operating system & updates

• **Set a minimum-OS compliance policy** (macOS 14+ recommended) so Platform SSO works reliably and out-of-support Macs are flagged.

• **Manage updates with Intune** (declarative software updates) so the fleet stays patched without user intervention.

Rollout discipline

• **Pilot first, expand in waves.** Validate each wave with `ADEMAC-Validation.ps1` before widening.

• **Hide noisy Setup Assistant screens** for a clean, fast first-run, but keep privacy/location prompts that your policy requires.

• **Pre-stage apps as Required** (starting with Company Portal) so a freshly enrolled Mac is useful within minutes.

Operations & documentation

• **Keep a runbook** of the Apple ID, ABM org, MDM server name, token/cert expiry dates, and the SSO profile identifiers — the values that, if lost, are painful to recover.

• **Report monthly** with `ADEMAC-DeviceReport.ps1`: enrolled macOS count, ADE vs. other, OS spread, Platform SSO registration coverage, and compliance.

• **Test OS upgrades** against the SSO extension before broad deployment; Apple occasionally changes Platform SSO behavior between major releases.

Security posture

• **Combine with disk encryption** (FileVault) and a **macOS security baseline/compliance policy** for a complete managed posture (separate solutions cover these).

• **Limit local admin** and use Intune to manage privileged access; supervised + locked enrollment prevents users from quietly removing management.

8. Deliverables Summary — macOS Automated Device Enrollment with SSO

Everything included in this solution package, and what each piece is for.

Documentation (this guide)

• **Executive Summary** — the business case and outcome in one page.

• **Solution Overview** — what ADE and Platform SSO are, how they work together, the authentication methods, benefits, and honest limitations.

• **Requirements Guide** — every Apple/Intune prerequisite with how to **verify** and how to **implement if missing**, plus the category-level Global Requirements.

• **Implementation Guide** — exact, ordered, click-by-click steps across four phases: connect Apple to Intune, assign devices and define the enrollment experience, deploy Company Portal + Platform SSO, then pilot/validate/expand.

• **Testing & Validation** — eight tests with explicit pass/fail gates and an acceptance gate.

• **Troubleshooting** — top-down diagnosis for enrollment and Platform SSO failures.

• **Best Practices** — identity, enrollment hygiene, the all-important token/certificate lifecycle, OS/updates, rollout discipline, and security posture.

PowerShell scripts (Microsoft Graph) — `ADEMAC-Scripts.zip`

All scripts are tenant-side (run from an admin workstation with the Microsoft Graph PowerShell SDK); each has a header explaining **Purpose · How to run · Expected results**.

1. `ADEMAC-Discovery.ps1` — tenant readiness: APNs certificate status/expiry, ADE/enrollment token status/expiry, macOS enrollment profiles, Company Portal assignment, and Platform SSO profile presence. Flags anything expiring within 30 days.

1. `ADEMAC-Validation.ps1` — validates a specific Mac: enrolled, supervised, ADE, SSO profile applied, Entra-registered, and compliant — with a clear PASS/FAIL summary.

1. `ADEMAC-DeviceReport.ps1` — tenant-wide CSV report of macOS devices: enrollment type, OS version, ownership/supervision, Platform SSO registration state, and compliance.

1. `ADEMAC-Remediation.ps1` — safe remediation: re-sync the ADE token, re-assign the

enrollment/SSO profiles to a target group, and surface tokens/certs nearing expiry. Read-only by default; changes require an explicit confirmation switch.

1. `Set-PlatformSSOProfile.ps1` — creates and assigns the Platform SSO (SSO app extension) configuration profile with the correct Microsoft Enterprise SSO identifiers, URLs, and Secure Enclave method.

Hero image

- `ADEMAC-hero.png` — the branded product image (1200×1200) used on the marketplace listing.

How it's delivered

A branded **Word** document and **PDF** (cover, table of contents, page numbers, and a scripts appendix), plus all five PowerShell scripts in a single **ZIP** — yours to keep and reuse across **unlimited devices**.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

ADEMAC-DeviceReport.ps1

Purpose: Exports every macOS device in Intune with the fields that matter for this solution: enrollment type (ADE vs. other), OS version, ownership, supervision, compliance, last check-in, and primary user. Use it to measure ADE coverage and Platform SSO rollout.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\ADEMAC-DeviceReport.ps1 .\ADEMAC-DeviceReport.ps1 -CsvPath C:\Temp\macos-fleet.csv

ADEMAC-Discovery.ps1

Purpose: Confirms the Intune tenant is ready to auto-enroll Macs and register Platform SSO: Apple MDM Push (APNs) certificate, Automated Device Enrollment (ADE) token, macOS enrollment profile(s), Company Portal assignment, and a Platform SSO (SSO app extension) configuration profile. Flags anything expiring within 30 days.

How to run: Install the SDK once: Install-Module Microsoft.Graph -Scope CurrentUser Then: .\ADEMAC-Discovery.ps1 Reads only — makes no changes.

ADEMAC-Remediation.ps1

Purpose: Performs the common fixes that unblock ADE/SSO: re-sync the ADE token with Apple, surface tokens/certificates nearing expiry, and (optionally) re-assign the macOS enrollment profile and Platform SSO configuration profile to a target group.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once # Read-only audit (default — makes NO changes): .\ADEMAC-Remediation.ps1 # Trigger an ADE token sync and re-assignments (explicit opt-in): .\ADEMAC-Remediation.ps1 -Apply -GroupId <entra-group-object-id> -SsoPolicyId <id> -EnrollmentProfileId <id>

ADEMAC-Validation.ps1

Purpose: For one macOS device (by serial number or device name), confirms it is: enrolled in Intune, ADE/DEP-enrolled, supervised, corporate-owned, compliant, and that a Platform SSO configuration profile is applied. Produces a clear PASS/FAIL verdict.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\ADEMAC-Validation.ps1 -Serial C02XXXXXXXX .\ADEMAC-Validation.ps1 -DeviceName "Jane-MacBook"

Set-PlatformSSOProfile.ps1

Purpose: Provisions the Extensible Single Sign-On (SSO app extension / Platform SSO) configuration profile for macOS with the correct Microsoft Enterprise SSO identifiers, login URLs, and the Secure Enclave authentication method — the same profile described in the Implementation Guide §6/Step 11 — then assigns it to a target Entra group.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once # Preview only (no change): .\Set-PlatformSSOProfile.ps1 -GroupId <entra-group-object-id> -WhatIf # Create & assign:

```
.\Set-PlatformSSOProfile.ps1 -GroupId <entra-group-object-id> # Choose a different method:
.\Set-PlatformSSOProfile.ps1 -GroupId <id> -AuthMethod passwordSync
```